

Do Imports Reduce GDP?

First Year Macro

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Do Imports Reduce GDP?

1. The Problem
2. *Intuition*: Moonbase Macro
3. *Algebra*: National Accounting
4. *Economics*: Indirect Effects
5. Takeaways, Practice Questions / Exercises

A claim you'll hear in the news

"The trade deficit subtracted three points from GDP growth."

Explicit motivation for post-2015 trade wars: "imports lower GDP"

Sounds simple: imports enter the GDP formula with a minus sign, so more imports $\Rightarrow$ less GDP.

Today: is this true?



Do Imports Reduce GDP?

National accounting GDP identity:

$$Y = C + G + I + (X - M)$$

The simple reading:

$$\frac{\partial Y}{\partial M} = -1$$

- At first glance seems OK... it's right there, $-M$

Question: is M really independent of the other terms?

An Example: The Moonbase



Imagine you are a pioneering astronaut:

- you are on a moonbase that **produces nothing** (and does no work).
- **100% dependent upon imports from Earth.**
- Imports are £500 Million, Consumption is £500 Million, GDP is £0.
- **If imports fall** to £300 Million, Consumption falls to £300 Million
- **GDP is still £0.** We produce nothing (no buildings, no new data, etc).

Do Imports Reduce GDP?

Gross Domestic Product: Sum of spending on final goods and services produced domestically

$$(C^{dom} + C^{for}) + (G^{dom} + G^{for}) + (I^{dom} + I^{for}) + X^{dom} - M = Y$$

We can **group by location**

$$(C^{dom} + G^{dom} + I^{dom} + X^{dom}) + (C^{for} + G^{for} + I^{for}) - M = Y$$

But wait... M is foreign-produced goods...

$$(C^{for} + G^{for} + I^{for}) = M$$

So really **GDP** is actually:

$$\checkmark \quad Y = (C^{dom} + G^{dom} + I^{dom} + X^{dom}) + [M - M]$$

M , by definition, is not part of Y ! but that's **not the whole story**

$$\times \quad \frac{dY}{dM} \neq [+1 - 1]$$

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Why write the GDP identity this way?

Why do textbooks write $Y = C + I + G + (X - M)$ at all?

- **Easier to measure:** Statistical agencies observe total C, G, I, X and M
- **Harder:** “how much of consumption was imported”

The cost: the formula *looks* like M is subtracted, when its true net effect is zero.

Back to the Moonbase: Indirect Effects

Imagine you are a pioneering astronaut:

Imports from Earth are half (food/medicine) and half (tools/ materials):

- $M = 500$
- $C^{for} = 250$
- $I^{for} = 250$ (the change in the value of your assets: +tools +mats)

With the tools and materials, you build a new base module, valued at 400

- Domestic value-added (work done) is (400-250): $I^{dom} = 150$
- $Y = C^{dom} + G^{dom} + I^{dom} + X^{dom} = 0 + 0 + 150 + 0 = 150$
- **Now imports have boosted GDP!**

Do imports affect GDP? Yes — through $C^{dom}, G^{dom}, I^{dom}, X^{dom}$

Ultimately, the total effect depends on all domestic components

$$\frac{dY}{dM} = \frac{\partial C^{dom}}{\partial M} + \frac{\partial G^{dom}}{\partial M} + \frac{\partial I^{dom}}{\partial M} + \frac{\partial X^{dom}}{\partial M}$$

Two real channels through which imports and GDP *do* interact:

1. Substitution: [negative comovement with M] *E.g. A consumer buys an imported car instead of a domestic one. GDP falls, not because the identity says so, but because a domestic sale no longer happens.*

2. Complementarity: [positive comovement with M] *E.g A domestic aircraft manufacturer needs a specialized imported component. Tariff the import, and the manufacturer can't get the part. Output (and GDP) falls with M.*

HH complementarities (I like to drink German beer on a UK-made sofa) or Firm substitution (the Chinese EV is more efficient than UK-made) are also possible

Discussion 1: Substitutes and Complements

A government imposes a large tariff on imported steel.

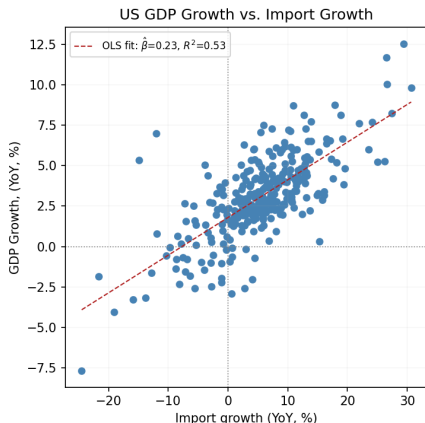
Question for you: does GDP go up or down? What would you need to know to answer confidently?

What if its not steel but Bananas or Coffee that can't be produced domestically?

Discussion 2: Going to the Data

Use the code provided to download US import and GDP growth from FRED

Task: Plot the time series and scatter plot. The naive correlation suggests imports actually boost GDP. Discuss why the naive correlation doesn't tell us which way causality runs



Discussion 3: Pricing the Priceless

In the moonbase-investment example, we priced the new hab at 400

Can you see any problem with this? How can we justify this based on:

- Cost-of-inputs to produce the hab
- Imputed Market Value: The value of the structure back on earth

In reality a lot of economic measurement faces the problem of market-less goods: government services with no private sector equivalent: public education, defence, NHS, owner-occupied housing (homeowners both hold the asset and use its services)